

Outline

- 1 Motivation and background
- 2 Extended STS
- 3 Numerical results
- 4 Software and outlook

Outline

1 Motivation and background

2 Extended STS

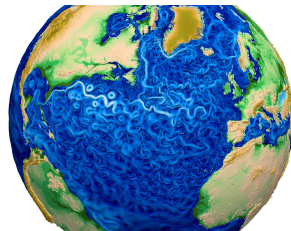
3 Numerical results

4 Software and outlook

Multiphysics simulations

Multiphysics models couple processes with distinct stability, locality, and cost.

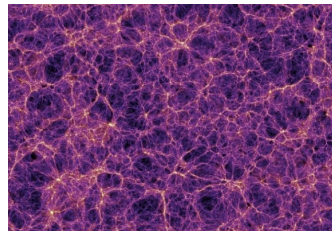
- Climate: transport, diffusion, chemistry, condensation, and exchange with ocean/ice/land.
- Astrophysics: fluid dynamics, gravity, radiation transport, and reaction networks.
- Plasma and combustion: spatial transport coupled to stiff, often local, source terms.



After spatial discretization,

$$\mathbf{y}'(t) = \sum_k \mathbf{f}^{\{k\}}(t, \mathbf{y}), \quad \mathbf{y}(t_0) = \mathbf{y}_0.$$

The partition is not cosmetic: each component wants a different numerical treatment.



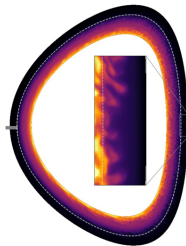
Target application: continuum gyrokinetic fusion simulations

[Francisquez et al., 2026]

The long-wavelength, full- f model evolves each species distribution $f_s(t, \mathbf{R}, v_{\parallel}, \mu)$:

$$\begin{aligned} \frac{\partial(\mathcal{J}f_s)}{\partial t} = & -\nabla_{\mathbf{R}} \cdot (\mathcal{J}\dot{\mathbf{R}}f_s) - \frac{\partial}{\partial v_{\parallel}}(\mathcal{J}\dot{v}_{\parallel}f_s) \\ & + \mathcal{J}\mathcal{C}_s^{\text{drag}}(f) + \mathcal{J}(\mathcal{S}_s + \mathcal{D}_s) + \mathcal{J}\mathcal{C}_s^{\text{diff}}(f). \end{aligned}$$

- Dougherty operator $\mathcal{J}\mathcal{C}_s^{\text{diff}}(f)$ combines velocity-space drag & diffusion; conserves particles, momentum, and energy.
- Imposes a parabolic time-step restriction $\Delta t \propto \Delta v^2$.



Design target

Explicitly evolve phase-space transport & collisional drag, and incorporate velocity-space diffusion—ideally without a globally coupled implicit phase-space solve.

Discretization in Gkeyll

- Conservative modal DG on mapped configuration-velocity grids.
- Hybrid $p = 1$ basis: linear in configuration space and μ , quadratic in $v_{\parallel}$.
- Full 3x2v capability with MPI and CUDA GPU support.

One model, three numerical personalities

We consider advection–diffusion–reaction systems

$$\mathbf{y}'(t) = \mathbf{f}^A(t, \mathbf{y}) + \mathbf{f}^D(t, \mathbf{y}) + \mathbf{f}^R(t, \mathbf{y}).$$

component	spectral structure	natural treatment	computational character
advection	mostly imaginary	explicit RK	local/neighbor communication
diffusion	negative real axis	STS	explicit, many cheap stages
reaction	stiff, often local	L -stable DIRK	local nonlinear solves

Design objective

Couple the three treatments accurately and adaptively, without turning diffusion into a globally coupled implicit solve.

Super-time-stepping: stabilized explicit diffusion

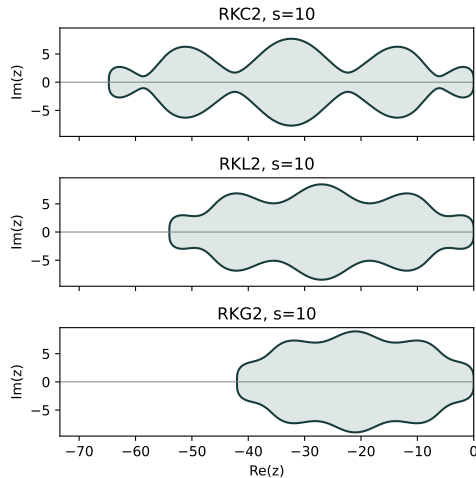
[Sommeijer et al., 1998; Meyer et al., 2014; Skaras et al., 2021]

For $y' = f(t, y)$, with $\lambda(f_y) \subset \mathbb{R}^-$, s -stage STS methods provide:

- an explicit, low-storage stage recurrence;
- real stability interval with extent $\mathcal{O}(s^2)$; and
- current families RKC [1998], RKL [2014], RKG [2021]—all second order.

Convex monotone property [Meyer et al., 2014]

RKL and RKG can preserve monotonicity for scalar parabolic problems throughout their admissible diffusion range; RKC generally does not.



STS trades additional local work for a much larger stable diffusion step.

Algorithmic structure of one STS step

[Sommeijer et al., 1998; Meyer et al., 2014]

For $\mathbf{y}' = \mathbf{f}(t, \mathbf{y})$ over a step of size h :

$$\mathbf{z}_0 = \mathbf{y}_n,$$

$$\mathbf{z}_1 = \mathbf{z}_0 + h\tilde{\mu}_1 \mathbf{f}(t_n, \mathbf{z}_0),$$

$$\begin{aligned} \mathbf{z}_j = & \mu_j \mathbf{z}_{j-1} + \nu_j \mathbf{z}_{j-2} + (1 - \mu_j - \nu_j) \mathbf{z}_0 \\ & + h\tilde{\mu}_j \mathbf{f}(T_{j-1}, \mathbf{z}_{j-1}) + h\tilde{\gamma}_j \mathbf{f}(t_n, \mathbf{z}_0), \quad j = 2, \dots, s, \end{aligned}$$

$$\mathbf{y}_{n+1} = \mathbf{z}_s.$$

Here $T_j = t_n + hc_j$, where c_j is the j th stage abscissa.

- The stage count s is chosen from the diffusion spectral radius.
- Coefficients derived from recursion relations for Chebyshev, Legendre, or Gegenbauer polynomials.
- Limited memory footprint: 5 vectors suffice for the recurrence.

STS handles diffusion alone; the question is how to couple it with advection and reaction.

Implicit-explicit additive Runge–Kutta methods

[Ascher et al., 1997; Kennedy & Carpenter, 2003]

For $\mathbf{y}' = \mathbf{f}^E + \mathbf{f}^I$, combine explicit and diagonally implicit RK tables:

$$\begin{aligned}\mathbf{z}_i &= \mathbf{y}_n + h \sum_{j=1}^{i-1} a_{ij}^E \mathbf{f}^E(T_j, \mathbf{z}_j) + h \sum_{j=1}^i a_{ij}^I \mathbf{f}^I(T_j, \mathbf{z}_j), \quad i = 1, \dots, s, \\ \mathbf{y}_{n+1} &= \mathbf{y}_n + h \sum_{j=1}^s \left[b_j^E \mathbf{f}^E(T_j, \mathbf{z}_j) + b_j^I \mathbf{f}^I(T_j, \mathbf{z}_j) \right], \\ \tilde{\mathbf{y}}_{n+1} &= \mathbf{y}_n + h \sum_{j=1}^s \left[\tilde{b}_j^E \mathbf{f}^E(T_j, \mathbf{z}_j) + \tilde{b}_j^I \mathbf{f}^I(T_j, \mathbf{z}_j) \right].\end{aligned}$$

Here $T_j = t_n + c_j h$, where the shared abscissae satisfy $c = A^E \mathbf{1} = A^I \mathbf{1}$.

- High-order coupling and an embedded error estimate come naturally.
- For ADR, the usual split is $\mathbf{f}^E = \mathbf{f}^A$ and $\mathbf{f}^I = \mathbf{f}^D + \mathbf{f}^R$.
- Each implicit stage then couples diffusion across the full spatial domain, even when reactions are local.

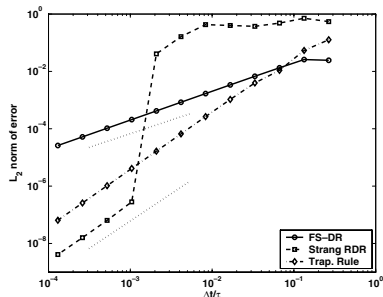
Operator splitting: flexible but loosely coupled

[Strang, 1968; Ropp & Shadid, 2005]

A Strang–Marchuk step composes independently evolved component problems:

$$\mathbf{y}_n \xrightarrow{\mathbf{f}^A, h/2} \mathbf{z}_1 \xrightarrow{\mathbf{f}^D, h/2} \mathbf{z}_2 \xrightarrow{\mathbf{f}^R, h} \mathbf{z}_3 \xrightarrow{\mathbf{f}^D, h/2} \mathbf{z}_4 \xrightarrow{\mathbf{f}^A, h/2} \mathbf{y}_{n+1}.$$

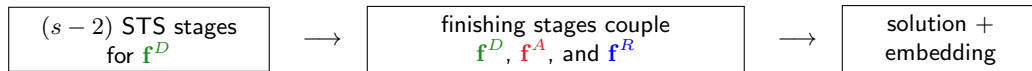
- Each component may use its ideal solver: ERK, STS, or DIRK.
- This Strang–Marchuk composition is second order, regardless of higher sub-solver orders.
- Higher order combinations are possible, but require backward time integration.
- Standard embedded temporal error estimates are unavailable.



Splitting error can dominate component-solver error.

PIROCK: STS diffusion with an ImEx finishing procedure

[Abdulle & Vilmart, 2013]



- Retains explicit stabilized diffusion and local implicit reaction solves.
- Includes an inexpensive temporal error estimate for adaptive stepping.
- The construction and coefficient tables are tied to the “ROCK2” STS method.
- Advection and reaction enter mainly after the diffusion stages, limiting coupling when operators strongly interact.

Closest comparator

PIROCK is the closest existing method to this work and the most direct numerical comparator.

Why existing choices leave a gap

Three established families, each giving up something multiphysics ADR needs:

ImEx ARK

Strong coupling and embedded error estimates.

Implicit $\mathbf{f}^D + \mathbf{f}^R$ creates global nonlinear systems.

Operator splitting

Each component uses its preferred solver.

Weak coupling, no inexpensive embeddings.

PIROCK

STS diffusion plus time adaptivity.

Tied to ROCK2, only applies $\mathbf{f}^A$, $\mathbf{f}^R$ at end of step.

The gap

Can we retain solver flexibility, embedded adaptivity, and stronger inter-operator coupling?

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- 2 **Extended STS**
- 3 Numerical results
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Multirate infinitesimal coupling: the transferable idea [Wensch et al., 2009; Sandu, 2019; Chinomona & R., 2021]

For a problem $\mathbf{y}'(t) = \mathbf{f}^E(t, \mathbf{y}) + \mathbf{f}^I(t, \mathbf{y}) + \mathbf{f}^F(t, \mathbf{y})$, MRI methods solve modified fast problems

$$\mathbf{v}'_i(t) = \mathbf{f}^F(t, \mathbf{v}_i) + \mathbf{r}_i(t), \quad \mathbf{v}_i(T_{i-1}) = \mathbf{z}_{i-1}, \quad t \in [T_{i-1}, T_i],$$

where $\mathbf{r}_i(t)$ combines already-computed values from $\mathbf{f}^E$ and $\mathbf{f}^I$ (next slide).

- Second-order explicit MIS methods follow from second-order stiffly-accurate RK tables with sorted abscissae, $0 = c_1 < c_2 < \dots < c_s = 1$.
- Implicit and ImEx variants additionally require a solve-decoupled implicit table.
- RK tables become coupling coefficients that encode how slow tendencies force each inner evolution.
- Higher-order MRI methods (up to $\mathcal{O}(h^6)$) are also available.

Note

The inner evolution need not represent a genuinely faster physical process. It can instead be a specialized solver for a structured operator.

Algorithmic structure of one MRI step

[Sandu, 2019; Chinomona & R., 2021]

- ❶ Set $\mathbf{z}_1 = \mathbf{y}_n$, $T_i = t_n + c_i h$, and $\Delta c_i = c_i - c_{i-1}$.
- ❷ For $i = 2, \dots, s$:

- If $\Delta c_i > 0$, define $\gamma_{ij}(t) = \sum_{k>0} \gamma_{ij}^{\{k\}} \left(\frac{t-T_{i-1}}{\Delta c_i} \right)^{k-1}$, $\omega_{ij}(t) = \sum_{k>0} \omega_{ij}^{\{k\}} \left(\frac{t-T_{i-1}}{\Delta c_i} \right)^{k-1}$,

$$\mathbf{r}_i(t) = \frac{1}{\Delta c_i} \left[\sum_{j=1}^{i-1} \gamma_{ij}(t) \mathbf{f}^I(T_j, \mathbf{z}_j) + \sum_{j=1}^{i-1} \omega_{ij}(t) \mathbf{f}^E(T_j, \mathbf{z}_j) \right], \quad \text{and} \quad \mathbf{v}_i(T_{i-1}) = \mathbf{z}_{i-1}.$$

Evolve $\mathbf{v}'_i = \mathbf{f}^F(t, \mathbf{v}_i) + \mathbf{r}_i(t)$ over $[T_{i-1}, T_i]$, and set $\mathbf{z}_i = \mathbf{v}_i(T_i)$.

- If $\Delta c_i = 0$, solve the implicit system for $\mathbf{z}_i$:

$$\mathbf{z}_i = \mathbf{z}_{i-1} + h \sum_{j=1}^i \bar{\gamma}_{ij} \mathbf{f}^I(T_j, \mathbf{z}_j) + h \sum_{j=1}^{i-1} \bar{\omega}_{ij} \mathbf{f}^E(T_j, \mathbf{z}_j),$$

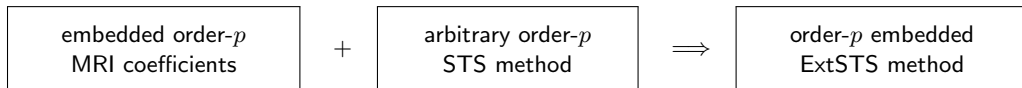
where $\bar{\gamma}_{ij} = \int_0^1 \gamma_{ij}(\tau) d\tau$ and $\bar{\omega}_{ij} = \int_0^1 \omega_{ij}(\tau) d\tau$.

- ❸ Set $\mathbf{y}_{n+1} = \mathbf{z}_s$; an extra coupling row emulates $i = s$ to produce $\tilde{\mathbf{y}}_{n+1}$.

Extended STS idea: replace the inner fast solve by one STS step for diffusion.

Extended STS construction in one picture

[R., Amihire & Aggul, 2026]



- $\mathbf{f}^F \rightarrow \mathbf{f}^D$: advanced by RKC, RKL, RKG, or another compatible STS solver.
- $\mathbf{f}^E \rightarrow \mathbf{f}^A$: explicit evaluations become forcing during the STS evolution.
- $\mathbf{f}^I \rightarrow \mathbf{f}^R$: explicit evaluations become forcing during the STS evolution; implicit solutions are performed during zero-length intervals.
- The embedded MRI formula supplies a temporal error estimate for adaptive stepping.

Note

This is a family and a construction recipe, not one fixed coefficient table.

Constructing an $\mathcal{O}(h^2)$ extended STS stage from an ImEx-ARK table

Given an s -stage ImEx-ARK method with Butcher tables (A^E, b^E) and (A^I, b^I) , where $c = A^E \mathbf{1} = A^I \mathbf{1}$ satisfies $0 = c_1 \leq \dots \leq c_s = 1$ and $a_{ii}^I \Delta c_i = 0$, we define the vector

$$\mathbf{g}_i = h \sum_{j=1}^{i-1} \left[(a_{ij}^I - a_{i-1,j}^I) \mathbf{f}^R(T_j, \mathbf{z}_j) + (a_{ij}^E - a_{i-1,j}^E) \mathbf{f}^A(T_j, \mathbf{z}_j) \right].$$

If $\Delta c_i > 0$: STS evolution

$$\mathbf{v}'_i = \mathbf{f}^D(t, \mathbf{v}_i) + \frac{\mathbf{g}_i}{\Delta c_i h}, \quad t \in [T_{i-1}, T_i],$$

with $\mathbf{v}_i(T_{i-1}) = \mathbf{z}_{i-1}$, then $\mathbf{z}_i = \mathbf{v}_i(T_i)$.

If $\Delta c_i = 0$: reaction solve

$$\mathbf{z}_i - h a_{ii}^I \mathbf{f}^R(T_i, \mathbf{z}_i) = \mathbf{z}_{i-1} + \mathbf{g}_i.$$

Only $\mathbf{f}^R$ is implicit; for local chemistry these solves decouple spatially.

Every stage is either an explicit STS evolution or a local implicit solve — never both.

MRI theory inherited by extended STS methods

$\mathcal{O}(h^2)$ ImEx–MRI theorem [adapted from Chinomona & R., 2021]

Consider an ImEx–MRI method constructed from an ImEx–ARK table with order at least two. If the coupling is *internally consistent*, i.e.,

$$\Gamma^{\{1\}} \mathbf{1} = \Omega^{\{1\}} \mathbf{1} = \Delta c, \quad \Gamma^{\{k\}} \mathbf{1} = \Omega^{\{k\}} \mathbf{1} = 0, \quad k > 1.$$

and the inner method has order $p \geq 2$, then the resulting MRI method is $\mathcal{O}(h^2)$.

More generally, $\mathcal{O}(h^p)$ requires both:

- 1 base methods and MRI coupling conditions through order p , and
- 2 an inner solver of order at least p .

Note

MRI method accuracy depends only on the $\Gamma^{\{k\}}$ and $\Omega^{\{k\}}$, and the inner solver's order, not on how that solver is implemented. We currently have MRI methods up to $\mathcal{O}(h^6)$; ExtSTS methods are currently limited to $\mathcal{O}(h^2)$ due to the STS sub-solver.

Why the extended STS construction is second order

Given ImEx-ARK tables (A^E, b^E) and (A^I, b^I) that

- each have order two and share $0 = c_1 \leq \dots \leq c_s = 1$;
- are stiffly accurate; and
- are solve-decoupled: $a_{ii}^I \Delta c_i = 0$ for every stage.

Writing the ExtSTS g_i in terms of MRI coefficients,

$$\gamma_{ij}^{\{1\}} = a_{ij}^I - a_{i-1,j}^I, \quad \omega_{ij}^{\{1\}} = a_{ij}^E - a_{i-1,j}^E, \quad \text{and} \quad \gamma_{ij}^{\{k\}} = \omega_{ij}^{\{k\}} = 0, \quad k > 1.$$

Summing and leveraging ImEx-ARK internal consistency:

$$\sum_j \gamma_{ij}^{\{1\}} = c_i - c_{i-1} = \Delta c_i, \quad \sum_j \omega_{ij}^{\{1\}} = \Delta c_i.$$

Result

ImEx-MRI internal consistency is thus satisfied, so ExtSTS methods are second order when the STS inner solver is also second order.

The same argument covers explicit and implicit variants.

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“Brusselator” benchmark tests

[Prigogine & Nicolis, 1967]

We compare ExtSTS against ImEx ARK, Strang–Marchuk, and PIROCK on representative test problems:

1D three-species ADR system on $\Omega = [0, 1]$

$$u_t = -cu_x + du_{xx} + r(A - (w + 1)u + vu^2),$$

$$v_t = -cv_x + dv_{xx} + r(wu - vu^2),$$

$$w_t = -cw_x + dw_{xx} + r((B - w)/\varepsilon - wu).$$

512 grid points; $A = 1$, $B = 3$, $c = 0.5$, $r = 1$; periodic or stationary boundaries; $d \in \{0.01, 0.1\}$ and $\varepsilon \in \{10^{-2}, 10^{-4}, 10^{-6}\}$.

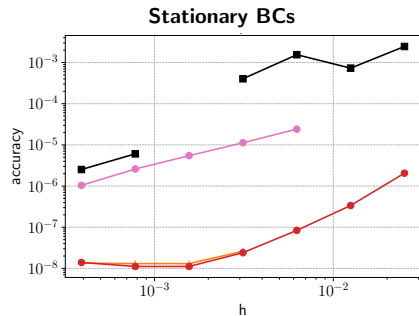
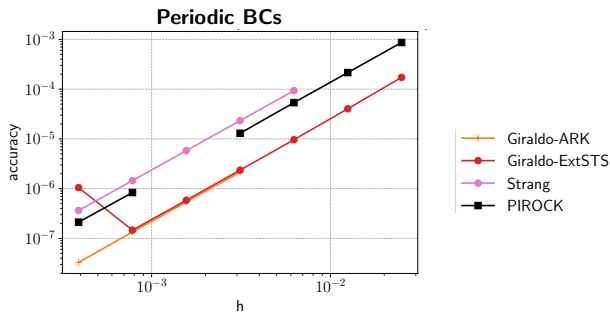
2D advective Brusselator on $\Omega = [0, 1]^2$

$$u_t = d\nabla^2 u + \mathbf{c}_u \cdot \nabla u + A + u^2 v - (B + 1)u,$$

$$v_t = d\nabla^2 v + \mathbf{c}_v \cdot \nabla v + Bu - u^2 v.$$

400×400 grid; periodic or stationary boundaries; $d \in \{0.01, 0.1\}$ and $B \in \{3, 30, 300\}$.
 $\mathbf{f}^A + \mathbf{f}^R$ is explicit (as done in PIROCK paper).

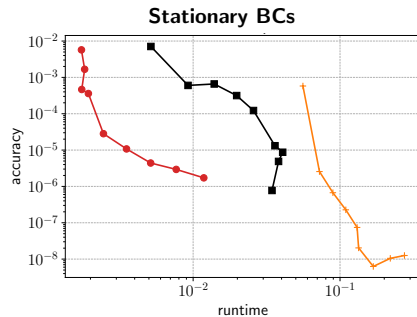
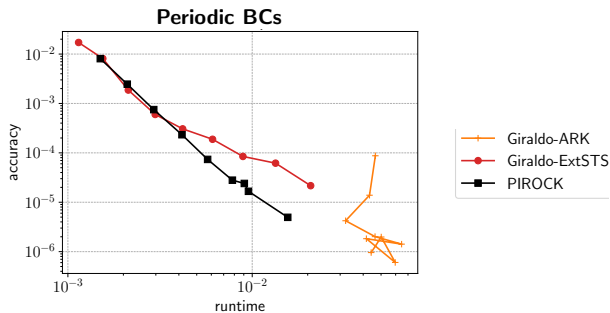
Fixed steps in 1D reveal the coupling deficiency in Strang and PIROCK



Convergence ($d = 10^{-2}$, $\varepsilon = 10^{-4}$): relative solution error versus step size; missing points indicate failed runs.

- ExtSTS is the only family robust at every requested fixed step size.
- ARK and Strang are generally unstable at the largest step sizes; PIROCK also fails at some intermediate step sizes.
- Stationary boundaries expose substantially larger errors in the weakly coupled methods.

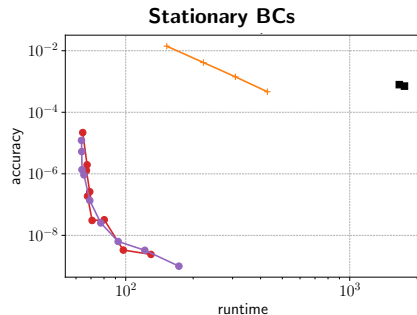
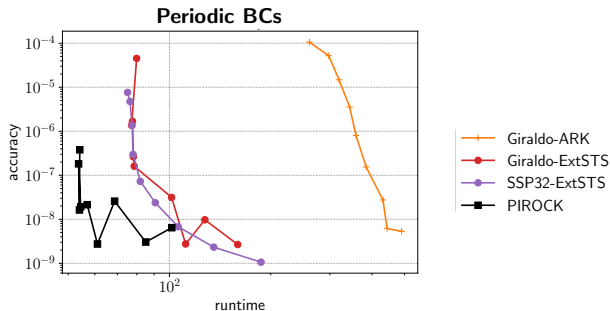
Adaptive stepping in 1D converts robustness into efficiency



Adaptive runtime efficiency ($d = 10^{-2}$, $\varepsilon = 10^{-4}$): relative solution error versus wall-clock time.

- Periodic case: PIROCK can be slightly more efficient for tighter tolerances.
- Stationary case: advection and diffusion oppose at the downwind boundary; tighter coupling makes ExtSTS efficient, while PIROCK deteriorates.
- Coupled $\mathbf{f}^D + \mathbf{f}^R$ causes ARK to lag behind both ExtSTS and PIROCK.

2D ADR tests tell the same coupling story



Adaptive runtime efficiency ($d = 0.1$, $B = 300$): relative solution error versus wall-clock time.

- Periodic: PIROCK leads, followed closely by ExtSTS; all STS-based methods beat ARK.
- Stationary: both ExtSTS variants are substantially more efficient/robust than PIROCK and ARK.
- SSP32-ExtSTS is competitive with Giraldo-ExtSTS.

Most efficient adaptive method across all test configurations

test	parameters	periodic BCs	stationary BCs
1D ADR	$d = 0.01, \varepsilon = 10^{-4}$	PIROCK (slightly)	ExtSTS
1D ADR	$d = 0.1, \varepsilon = 10^{-2}$	PIROCK	ExtSTS
1D AD	$d = 0.01$	Giraldo-ExtSTS	Giraldo-ExtSTS
1D AD	$d = 0.1$	PIROCK	Giraldo-ExtSTS
1D RD	$d = 0.1, \varepsilon \in \{10^{-2}, 10^{-4}\}$	PIROCK (ExtSTS close)	ExtSTS
2D ADR	$d = 0.01, B = 3$	Giraldo-ExtSTS	ExtSTS (both similar)
2D ADR	$d = 0.01, B = 30$	PIROCK loose / Giraldo-ExtSTS tight	ExtSTS (both similar)
2D ADR	$d = 0.01, B = 300$	PIROCK loose / Giraldo-ExtSTS tight	SSP32-ExtSTS
2D ADR	$d = 0.1, B \in \{3, 30, 300\}$	PIROCK (ExtSTS close)	Giraldo-ExtSTS

- ExtSTS is the most efficient method or the runner-up in every configuration.
- PIROCK is never the most efficient with stationary boundaries.
- ARK is never the most efficient: global implicit diffusion is too costly.

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SUNDIALS [Hindmarsh et al., *TOMS*, 2005; Balos et al., *ParComp*, 2021; Gardner et al., *TOMS*, 2022; Roberts et al., *TOMS*, 2026]

One of my primary goals is to impact other fields – I choose to achieve this through dissemination of modern methods through high-quality open-source software.

My collaborators and I develop and maintain *SUNDIALS* – the Suite of Nonlinear and Differential/ALgebraic Solvers

- Adaptive time integrators for ODEs and DAEs and efficient nonlinear solvers, used throughout research and industry.
- Written in C/C++ (w/ Python and Fortran interfaces).
- Native parallel support: GPU, MPI, and threading.
- Runs anywhere from laptops to supercomputers.
- BSD license; available from GitHub or Spack.
- Over 100,000 downloads/year since 2016.
- Embedded in MATLAB, Mathematica, Python (scikit), DifferentialEquations.jl, OpenModelica, ...
- CVODE(S): linear multistep ODE methods, fwd/adj sensitivities
- IDA(S): linear multistep DAE methods, fwd/adj sensitivities
- ARKODE: multi-stage ODE methods
- KINSOL: nonlinear solver



ExtSTS in SUNDIALS ARKODE

[R. et al., 2023; Roberts et al., 2026]

ARKODE has been part of SUNDIALS since 2014. Starting with adaptive ImEx-ARK methods, it now includes a variety of composable time-stepping modules:

- ARKStep: ERK, DIRK, and ImEx ARK,
- ERKStep: ERK, and fixed-step discrete FSA/ASA,
- MRISStep: MIS, MRI-GARK, and related methods,
- LSRKStep: low-storage explicit SSP and STS methods,
- SPRKStep: fixed-step symplectic RK methods for partitioned Hamiltonian systems,
- SplittingStep: fixed-step operator splitting methods (including very high order).

Available now

ExtSTS methods were released in
SUNDIALS v7.9.0 (September 2026).

- adaptive ImEx, explicit, and implicit variants
- interchangeable RKC/RKL diffusion steppers (RKG coming soon)

Code: github.com/LLNL/sundials

Docs: sundials.readthedocs.io

Take-away message: Extended STS fills a method gap

- Combines **STS diffusion**, **explicit transport**, and **local implicit reactions** within a tightly coupled and temporally adaptive MRI framework.
- The construction is modular: existing ARK/MRI coefficients can be paired with RKC, RKL, or related STS methods.
- ExtSTS inherits MRI order theory: order- p coupling plus an order- p STS solver gives order p . The proposed methods are second order, solve-decoupled, and embedded; the construction is not limited to second order—the STS families used here are.
- Experiments show robust performance across dimensions, parameters, and boundary conditions—especially when operators strongly interact.

The practical message

Avoiding global implicit diffusion need not require weak operator coupling.

Next steps

- Apply ExtSTS to Gkeyll 5D gyrokinetics simulations on GPU-based architectures.
- Construct third- and fourth-order STS methods with analytic stage recurrences—the missing ingredient for higher-order ExtSTS.
- Develop multiphysics STS extensions that also subcycle genuinely faster physical scales.
- Compare production implementations against scalable implicit diffusion solvers on HPC systems.
- Determine when ExtSTS methods can inherit positivity, SSP, or invariant-domain-preserving properties from their base methods.

Longer-term goal

Make structure-exploiting time integration as composable and portable as the spatial discretizations it advances.

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CEDA: *Computational Evaluation and Design of Actuators for Core-Edge Integration*



FASTMath: *Frameworks, Algorithms and Software Technologies for Mathematics*

Questions?

Backup slides

Numerical results

- Periodic vs. stationary BCs
- Experimental protocol
- RKC vs. RKL sub-solvers
- 1D advection–diffusion
- 1D reaction–diffusion

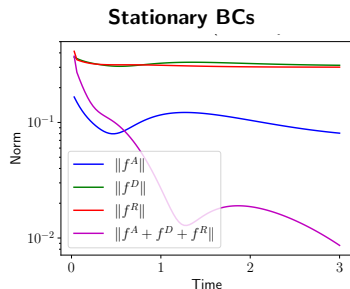
Method construction

- Joint stability function
- Joint stability regions
- Padding Giraldo's ARK2
- SSP32-ExtSTS
- Embedded error estimation

Coupling

- Commutators and splitting error

Periodic BCs Stationary



$\|\mathbf{f}^A\|$ and $\|\mathbf{f}^D\|$ decay as the solution flattens; the total tendency is essen-

- Stationary: fixed boundary values sustain gradients, so all three operators stay large while their sum is up to an order of magnitude smaller.
- Methods that apply the operators in sequence (Strang; PIROCK's finishing stages) must recover this balance across sub-steps—consistent with their sensitivity to the boundary type.

Backup: experimental protocol

Methods

- **ExtSTS**: Giraldo (ImEx/explicit/implicit) and SSP32; RKL diffusion sub-solver.
- **ARK**: unpadded Giraldo tables; $\mathbf{f}^A$ explicit, $\mathbf{f}^D + \mathbf{f}^R$ implicit.
- **Strang–Marchuk** (fixed steps only): RKL for $\mathbf{f}^D$, Giraldo ARK for $\mathbf{f}^A + \mathbf{f}^R$.
- **PIROCK**: authors' Fortran-77 code (ROCK2 diffusion), test problem modified to match; same partitioning and tolerances.

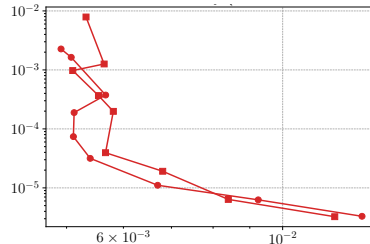
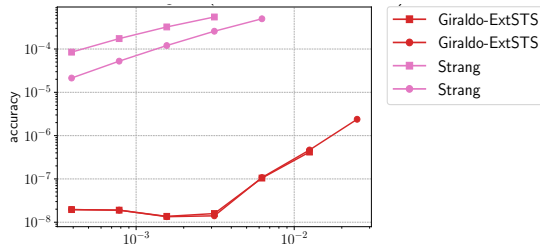
Settings

- Implicit solves: Newton with a direct linear solver (ExtSTS/Strang) or unpreconditioned GMRES (ARK).
- Adaptive: $\text{atol} = 10^{-11}$, $\text{rtol} = 10^{-2.5}, 10^{-3}, \dots, 10^{-6.5}$.
- Fixed (1D): $h = 1/40, 1/80, \dots, 1/5120$.
- Reference: ARK4(3)7L[2]SA₁ [Kennedy & Carpenter, 2019], $\text{rtol} = 10^{-10}$ (1D), 10^{-8} (2D).
- Space: second-order finite differences; 512 nodes on $t \in [0, 3]$ (1D), 400^2 on $t \in [0, 5]$ (2D).

All tests: github.com/sundials-codes/ceda-demonstrations

◀ Back to benchmarks

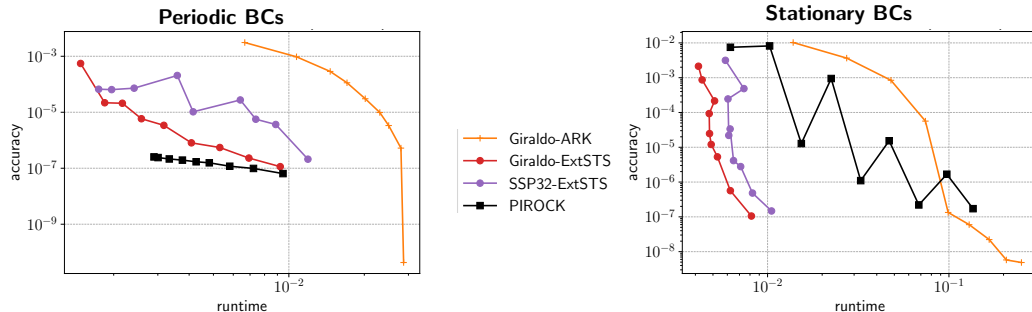
Backup: the STS sub-solver is genuinely modular for these tests



1D ADR, stationary BCs: fixed-step convergence (L) and adaptive efficiency (R). Squares: RKC; circles: RKL.

- ExtSTS convergence is insensitive to choice of RKC vs RKL.
- RKC's larger stability interval provides little advantage under adaptive stepping.
- Strang splitting is much more sensitive to the STS choice, and has much larger error constant.

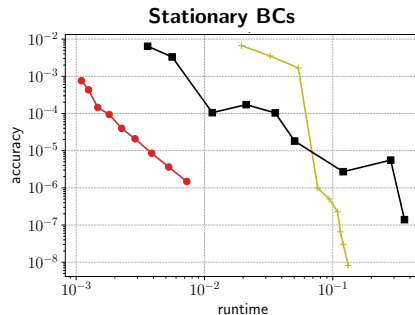
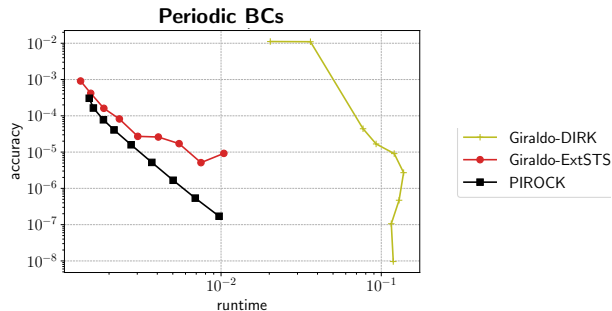
Backup: removing reactions in 1D widens the solve-cost gap



Adaptive runtime efficiency ($d = 0.1$): relative solution error versus wall-clock time.

- ExtSTS and PIROCK require no implicit solves for advection–diffusion.
- ARK still solves diffusion implicitly; its disadvantage grows with d .
- Giraldo-ExtSTS is consistently efficient across both boundary types, while PIROCK remains highly sensitive to the BC type.

Backup: 1D reaction–diffusion efficiency



Adaptive runtime efficiency ($d = 0.1$, $\varepsilon = 10^{-4}$): relative solution error versus wall-clock time.

- Periodic case: PIROCK is the most efficient method.
- Stationary case: ExtSTS is significantly more efficient and reaches the full requested accuracy range.
- The contrast again exposes the cost of weak coupling when the operators strongly interact.

Backup: stability must respect all three operators

[Giraldo et al., 2013; Fekete et al., 2022]

Analyze the partitioned test equation

$$y' = (\lambda^A + \lambda^D + \lambda^R)y, \quad \eta^\sigma = h\lambda^\sigma, \quad \lambda^D \in \mathbb{R}^-.$$

STS sub-solves evolve $v'_i = \eta^D v_i + \frac{g_i}{\Delta c_i h}$ over $[T_{i-1}, T_i]$ with $v_i(T_{i-1}) = z_{i-1}$ to give

$$z_i = \mathcal{R}_H^D(\Delta c_i \eta^D) z_{i-1} + \mathcal{R}_F^D(\Delta c_i \eta^D) g_i.$$

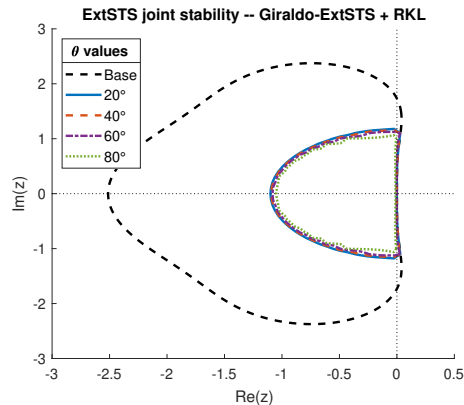
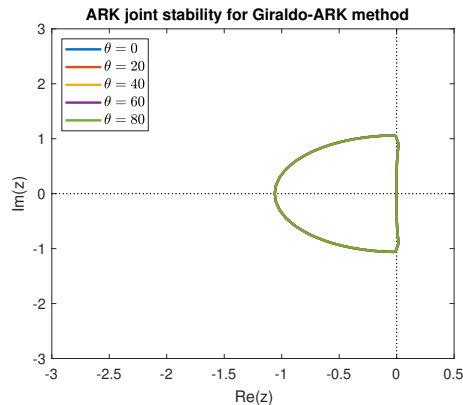
Stacking all stages gives the ExtSTS stability function

$$\mathcal{R}(\eta^D, \eta^A, \eta^R) = \mathbf{e}_s^T \left[I - P - Q(\eta^R A^I + \eta^A A^E) \right]^{-1} \mathbf{e}_1,$$

where P contains the homogeneous STS propagators and Q contains the forced propagators.

- Joint regions require $|\mathcal{R}| \leq 1$ for all $\eta^D \in [-\rho, 0)$ and all η^R in a wedge from $\mathbb{C}^-$.
- They isolate stability lost through coupling rather than through each component.
- RKC and RKL choices give nearly identical joint regions.
- Candidate families: Giraldo ImEx / explicit / implicit; embedded SSP(3,2).

Backup: representative joint stability regions



Experimentally for this and other base ARK, ERK, and DIRK tables

Extended STS methods approximately retain the underlying joint stability of their base RK methods, across a very large diffusion stiffness range.

Backup: padding Giraldo's ARK2 tables for ExtSTS

[Giraldo et al., 2013]

Original ImEx ARK ($\gamma = 1 - \frac{\sqrt{2}}{2}$, $\delta = \frac{\sqrt{2}}{4}$):

0	0	0	0 0	0	0	0 0
2γ	2γ	0	0 0	γ	γ	0 0
1	$\frac{3-2\sqrt{2}}{6}$	$\frac{3+2\sqrt{2}}{6}$	0 0	δ	δ	γ 0
1	δ	δ	γ 0	δ	δ	γ 0
	$\frac{4-\sqrt{2}}{8}$	$\frac{4-\sqrt{2}}{8}$	δ 0	$\frac{4-\sqrt{2}}{8}$	$\frac{4-\sqrt{2}}{8}$	δ 0

- Stages 2 and 3 have $a_{ii}^I = \gamma \neq 0$ with $\Delta c_i > 0$: an implicit solve over a nonzero interval, which ExtSTS forbids.
- Padding splits each into an STS evolution followed by a zero-length implicit solve.
- Zero columns 2 and 4: STS stage values never feed $\mathbf{f}^A$ or $\mathbf{f}^R$.

Padded, solve-decoupled, stiffly accurate tables ($c \mid A^E \mid A^I$):

0	0	0	0	0 0 0	0	0	0	0 0 0	
2γ	2γ	0	0	0 0 0	2γ	0	0	0 0 0	STS
2γ	2γ	0	0	0 0 0	γ	0	γ	0 0 0	solve
1	$\frac{3-2\sqrt{2}}{6}$	0	$\frac{3+2\sqrt{2}}{6}$	0 0 0	0	0	1	0 0 0	STS
1	$\frac{3-2\sqrt{2}}{6}$	0	$\frac{3+2\sqrt{2}}{6}$	0 0 0	δ	0	δ	0 γ 0	solve
1	δ	0	δ	0 γ 0	δ	0	δ	0 γ 0	explicit
	$\frac{4-\sqrt{2}}{8}$	0	$\frac{4-\sqrt{2}}{8}$	0 δ 0	$\frac{4-\sqrt{2}}{8}$	0	$\frac{4-\sqrt{2}}{8}$	0 δ 0	

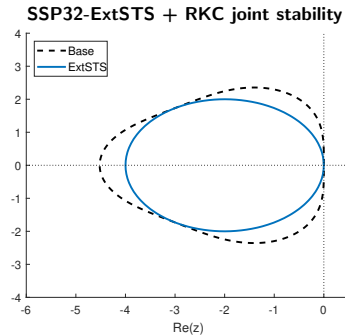
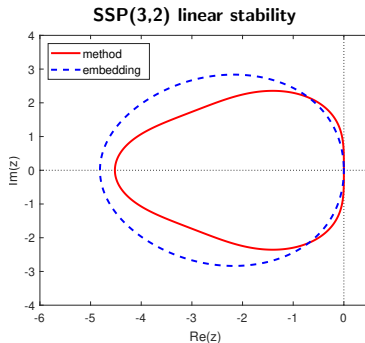
Orange entries are free, provided $c = A^I \mathbf{1}$ holds. The implicit table is A- and L-stable with stage order 1 and an A-stable embedding.

Backup: SSP32-ExtSTS for advection–diffusion

[Fekete et al., 2022]

Embedded SSP(3,2), in stiffly accurate form:

0	0	0	0	0
$\frac{1}{2}$	$\frac{1}{2}$	0	0	0
1	$\frac{1}{2}$	$\frac{1}{2}$	0	0
1	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0
	$\frac{4}{9}$	$\frac{1}{3}$	$\frac{2}{9}$	0



- The fourth row repeats b , making the table stiffly accurate as ExtSTS requires.
- With $\mathbf{f}^R = 0$ and diffusion stiffness up to $\rho = 10^6$, the joint region is only slightly smaller than the base region; RKL gives the same result.
- Intended for advection that needs SSP treatment (shocks, steep fronts); in our tests it trails Giraldo-ExtSTS slightly, except with the stiffest 2D reactions.
- Whether ExtSTS preserves the SSP property itself remains open.

Backup: embedded error estimation

Given embedding weights d^I, d^E , and writing $\mathbf{f}_j^R = \mathbf{f}^R(T_j, \mathbf{z}_j)$, $\mathbf{f}_j^A = \mathbf{f}^A(T_j, \mathbf{z}_j)$, form

$$\tilde{\mathbf{g}} = h \sum_{j=1}^s \left[(d_j^I - a_{s-1,j}^I) \mathbf{f}_j^R + (d_j^E - a_{s-1,j}^E) \mathbf{f}_j^A \right].$$

A second STS evolution over the final interval produces $\tilde{\mathbf{y}}_{n+1}$, and

$$\ell_{n+1} = \|\mathbf{y}_{n+1} - \tilde{\mathbf{y}}_{n+1}\|_{\text{WRMS}}$$

drives a standard adaptive step-size controller.

- No step-doubling is required.
- The embedding retains the same locality and solve-decoupled structure.

Backup: why weak coupling is sensitive to boundary conditions

For linear $\mathbf{y}' = (\mathbf{A} + \mathbf{D})\mathbf{y}$, with commutator $[X, Y] = XY - YX$, a Strang step satisfies

$$e^{\frac{\hbar}{2}A} e^{hD} e^{\frac{\hbar}{2}A} = \exp\left(h(A+D) - \frac{\hbar^3}{24}[A, [A, D]] - \frac{\hbar^3}{12}[D, [A, D]] + \mathcal{O}(\hbar^5)\right).$$

- Splitting error vanishes when $[A, D] = 0$; otherwise it grows with the stiffness of both operators.
- **Periodic, uniform grid, constant c, d :** A and D are circulant, so they commute and advection–diffusion splitting is exact.
- **Stationary BCs:** the boundary rows break translation invariance; $[A, D] \neq 0$ near the boundaries—where the operators balance (see previous backup).
- Reactions vary in space, so they introduce splitting error for both boundary types.

Interpretation (not analyzed in the paper)

ExtSTS feeds $\mathbf{f}^A$ and $\mathbf{f}^R$ into every STS sub-evolution as forcing, rather than alternating operators. We have not analyzed ExtSTS error constants; the claim that this reduces them rests on the numerical evidence.